

Loops: for, while, do-while, break, continue, patterns ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- Write for, while and do-while and compare them
- Use break and continue
- Print multiplication tables, factorial, nested-loop patterns

	for	while	do-while
Test	At the top	At the top	At the bottom — runs at least once
Best when	Count known	Condition, count unknown	Menus, "repeat until"
Header	for(s;c;u)	while(c)	do { } while(c);

```
/* table of n */
int n, i;
scanf("%d", &n);
for (i = 1; i ≤ 10; i++)
    printf("%d x %d = %d\n", n, i, n * i);
```

```
/* factorial : 4! = 24 */
int n, i; long f = 1;
scanf("%d", &n);
for (i = 1; i ≤ n; i++)
    f = f * i;
printf("%ld", f);
```

break vs continue

- break: leave the loop immediately.
- continue: skip the rest of this iteration and go to the next test / increment.

```
int i;
for (i = 1; i ≤ 5; i++) {
    if (i == 3) continue;
    printf("%d ", i);    /* 1 2 4 5 */
}
```

Nested loops — 2026 pattern

```
int i, j;
for (i = 1; i ≤ 5; i++) {
    for (j = 1; j ≤ i; j++)
        printf("%d", j);
    printf("\n");
}
/* 1
12
123
1234
12345 */
```

Number / Square: `for(i=1;i<=5;i++) printf("%d %d\n", i, i*i);`

HOW BIEK ASKS THIS

Programs to memorise: table, factorial, even 0–50, sum of n numbers, reverse digits, prime check, Fibonacci, patterns.

COMMON BOARD MISTAKE

do-while ends with a semicolon after `while(condition);` forgetting it is a syntax error. Infinite loop: `for(;;)` or `while(1)`.

اردو میں سمجھیں

for گنتی معلوم ہو تو، while شرط پر، do-while کم از کم ایک بار۔ break لوپ توڑتا ہے، continue اس چکر کو چھوڑتا ہے۔
ٹیبل، فیکٹوریل اور پیٹرن ہر سال آتے ہیں — زبانی لکھ کر یاد کریں۔

CHECK YOURSELF

1. $4! = ?$
2. Does do-while always run once?

Answers: 1. 24 · 2. Yes

MCQs

1. `for()` is used when we need:
 - a. Sequential structure
 - b. Selection structure
 - c. Iteration structure
 - d. Top down structure
2. This statement skips the rest of the loop body and continues from the top:
 - a. `break`
 - b. `resume`
 - c. `continue`
 - d. `skip`

Key: 1=c, 2=c

Short questions (3 marks each)

1. Write a program that prints the multiplication table of an inputted number.
2. Write a program that prints factorial of n (hint $4!=24$).
3. Differentiate while and do-while.
4. Define break and continue with an example.
5. Write a nested-loop program for the triangle 1 / 12 / 123 / ...

Model short answers

1. for i=1 to 10 print n x i = n*i.
2. f=1; for i=1..n f*=i; print f.
3. while tests first (may run 0 times). do-while tests after the body (at least once).
4. break exits; continue jumps to next iteration. Show a 4-line for-loop example.
5. Outer i=1..5, inner j=1..i print j, then newline.

Long questions (board Section C)

1. What is a loop? Explain for, while and do-while with programs. Discuss nested loops.